

Nges Brian, Njungle

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Education

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| Aug 2023 — Present | Ph.D. in Computer Engineering
Arizona State University, Tempe, Arizona, USA |
| Sept 2021 — April 2023 | Master of Business Administration
University of the People (UoP), Pasadena, California, USA |
| Sept 2014 — June 2018 | Bachelors of Engineering in Computer Engineering
University of Buea, South West Region, Cameroon, Africa |

Research Interests

Applied Cryptography— Formal Methods— Privacy-Preserving Computations — Hardware and Systems Security.
Build Scalable, performance-efficient secure systems that bridge security theories and real-world constraints.

Skills: Homomorphic Encryption, Zero-knowledge Proofs, Post-Quantum Cryptography, Formal Verification

Research and Academic Experiences

- **Adaptive & Secure Computing Systems (ASCS) Laboratory Lead**  Aug 2025 – Present
Secure, Trusted, and Assured Microelectronics (STAM) Center 
Recruit and mentor students, guide and support their research, and coordinate laboratory and center-wide activities. Identify funding opportunities and contribute to the preparation of grant proposals and white papers.
- **Graduate Researcher (Candidate) - Applied Cryptography** Advisor: Prof. Michel Kinsy Aug 2023 – Present
Conduct research on secure and privacy-preserving technologies, spanning advanced cryptographic theory, implementation, and systems integration. Focus areas include designing novel cryptographic algorithms and primitives, efficient application across heterogeneous computing domains, and formal verification of critical applications.

Relevant Research Projects and Tools

- **FHEON: Configurable Framework for Encrypted Neural Networks** ◦ Designed and implemented FHEON, the first of its kind configurable framework for encrypted neural networks training and inference.
 - Website: <https://fheon.pqcsecure.org> ◦ GitHub: <https://github.com/stamcenter/fheon>
- **SentinelTouch: Lightweight Privacy-Preserving Biometric-Fingerprinting Authentication system**
 - Introduced an efficient homomorphic encryption and neural networks biometric identification system. Achieved between $2.5\times$ to $10\times$ latency improvement ◦ GitHub: <https://github.com/stamcenter/sentineltouch>
- **FAVE: Constant-Time and Memory Safety Binary Analysis Tool** ◦ Proposed a new approach for constant-time verification based on low-level instruction traces. Designed and developed a binary analysis tool for side-channels and memory vulnerabilities detection in cryptographic and security-centric applications.
- **VIPER: Verifiable Private Information Retrieval** ◦ Designed and developed a remote database information retrieval system which enforces both privacy and data integrity in a non-interactive and computationally efficient manner. ◦ <https://www.pqcsecure.org/verifiable.html> ◦ <https://www.pqcsecure.org/pir.html>
- **Faster Crystals: High-Performance ML-KEM and ML-DSA** ◦ Optimized PQC primitives (ML-KEM and ML-DSA) for high-performance execution across all NIST FIPS security levels. ◦ Developed software and hardware-accelerated architectures, achieving up to 60% performance improvements on both referenced baselines.
- **End-to-End ZKP Accelerator:** ◦ Contributed to the design and develop of high-performance FPGA zero-knowledge proofs hardware primitives. Achieved between $10\times$ to $25\times$ speedups for different primitives.

Relevant Technical Work Experiences

- **Biosfera360**, Maryland, United States Jan 2023 – Aug 2023 / **Data Scientist (On-site)**
Designed and implemented machine learning models to analyze and predict desertification on geo-spatial trends. Technologies: Python, PyTorch, R, Computer Vision, Geo-statistics APIs, Google Earth and Maps
- **BlackHorse Solutions**, Virginia, USA Oct 2019 – Nov 2019 / **Software Engineer Intern (Remote)**
Prototyped quantum-safe cryptographic primitives and integrated them into security-centric systems. Technologies: C, C++, Wireguard VPN, Curve25519, ML-KEM and ML-DSA (FIPS 203 & FIPS 204)
- **ABEBOH LLC, Buea and Douala, Cameroon** Sept 2016 – Sept 2023 / **Co-Founder & CEO (On-site)**
Spearheaded the design and delivery of hundreds of client and in-house technology solutions. **Spotlight:** Led the development of a multi-award-winning music social media and marketplace serving 30,000+ users. Technologies: PHP, Python, Javascript, Yii2, Laravel, Django, Node.js, React JS, React Native, Wordpress

- **Google / Apache Software Foundation, USA** Mar 2016 – Aug 2016 / **GSoC Intern (Remote)**
Developed unit tests and the regression testing suite for the open-source Mynewt Operating System (IoT-OS)
Technologies: C and Arduino Programming.

Research Publications

Peer-Reviewed Conferences and Journals

- [1] **FHEON: A Configurable Framework for Developing Privacy-Preserving Neural Networks Using Homomorphic Encryption**, *Nges Brian Njungle*, Eric Jahns, Michel A. Kinsy. Proceedings on Privacy Enhancing Technologies Symposium (**PoPETs/PETs'26**), Issue 4, 2026.
- [2] **SentinelTouch: A Lightweight Privacy-Preserving Biometric-Fingerprinting Authentication and Identification System Based on Neural Networks and Homomorphic Encryption**, *Nges Brian Njungle*, Eric Jahns, Mishel Jyothis Paul, Michel A. Kinsy. Proceedings on Privacy Enhancing Technologies Symposium (**PoPETs/PETs'26**), Issue 2, 2026.
- [3] **PrivSpike: A Privacy-Preserving Inference Framework for Deep Spiking Neural Networks using Homomorphic Encryption**, *Nges Brian Njungle*, Eric Jahns, Milan Stojkov, Michel A. Kinsy. Applied Cryptography and Network Security (**ACNS'26**), Issue 1, 2026.
- [4] **Prismo: A Decision Support System for Privacy-Preserving ML Framework Selection**, *Nges Brian Njungle*, Eric Jahns, Luigi Mastromauro, Edwin P. Kayang, Milan Stojkov, Michel A. Kinsy. 22nd Annual International Conference on Privacy, Security, and Trust (**PST'26**),
- [5] **Gotta Hash 'Em All! Speeding Up Hash Functions for Zero-Knowledge Proof Applications**, Nojan Sheybani, Teng kai Gong, Anees Ahmed, *Nges Brian Njungle*, Michel A. Kinsy, Farinaz Koushanfar. Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (**ICCAD'25**), 2025.
- [6] **Activate Me!: Designing Efficient Activation Functions for Privacy-Preserving Machine Learning with Fully Homomorphic Encryption**, *Nges Brian Njungle*, Michel A. Kinsy. International Conference on Cryptology in Africa (**AFRICACRYPT'25**), Cham: Springer Nature Switzerland, 2025.
- [7] **A Safety-Centric Analysis and Benchmarks of Modern Open-Source Homomorphic Encryption Libraries**, *Nges Brian Njungle*, Milan Stojkov, Michel A. Kinsy. 22nd International Conference on Security and Cryptography (**SECRYPT'25**), Bilbao, Spain, 2025.
- [8] **GuardianML: Anatomy of Privacy-Preserving Machine Learning Techniques and Frameworks**, *Nges Brian Njungle*, Eric Jahns, Zhenqi Wu, Luigi Mastromauro, Michel A. Kinsy. **IEEE Access**, 2025.
- [9] **AMAZE: Accelerated MiMC Hardware Architecture for Zero-Knowledge on the Edge**, Anees Ahmed, Nojan Sheybani, Davi Moreno, *Nges Brian Njungle*, Teng kai Gong, Michel Kinsy, Farinaz Koushanfar. Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design (**ICCAD'24**) [***Top Picks in Hardware and Embedded Security 2025 Finalist***]

arXiv Preprints

- [1] **DALC-CT: Dynamic Analysis of Low-Level Code Traces for Constant-Time Verification**, *Nges Brian Njungle*, Edwin P. Kayang, Mishel J. Paul, Michel A. Kinsy. <https://arxiv.org/abs/2604.16832> 
- [2] **Toward Deep Encrypted Training: Low-Latency, Memory-Efficient, and High-Throughput Inference for Encrypted Neural Networks**, *Nges Brian Njungle*, Eric Jahns, Michel A. Kinsy. <https://arxiv.org/abs/2604.16834> 

Leadership, Awards & Others

Co-Instructor: CSE/CEN 598 Hardware Security & Trust	Spring 2026
Teaching Assistant: CSE 536 Advanced OS; CSE 240 Introduction to Programming Languages	Spring 2026
President , Muzikol Music Awards (MUMA),	2020 and 2021
Campus Adv. Council Mem. & Regional Ambassador Lead for Africa , Mozilla Rep., Mozilla	2015 – 2018
Shortlist , African Prize for Engineering and Innovation (Project: Muzikol)	2018
Pitch at Palace Africa (Project: Muzikol)	2018
Semi-Finalist , World Summit Award - Young Innovator Award (Project: Health Corner)	2018
Winner , Best Innovation at Bonteh Digital Media Awards (BDMA) (Project: Muzikol)	2017